

小狗充电桩SOP操作说明

文档包括以下四方面内容，具体如下：

1、充电桩固件烧录

小狗充电桩PCBA没有程序时，用JLINK烧录固件程序，若有程序，如果需要OTA更新程序，执行2、3.1和3.4步骤。

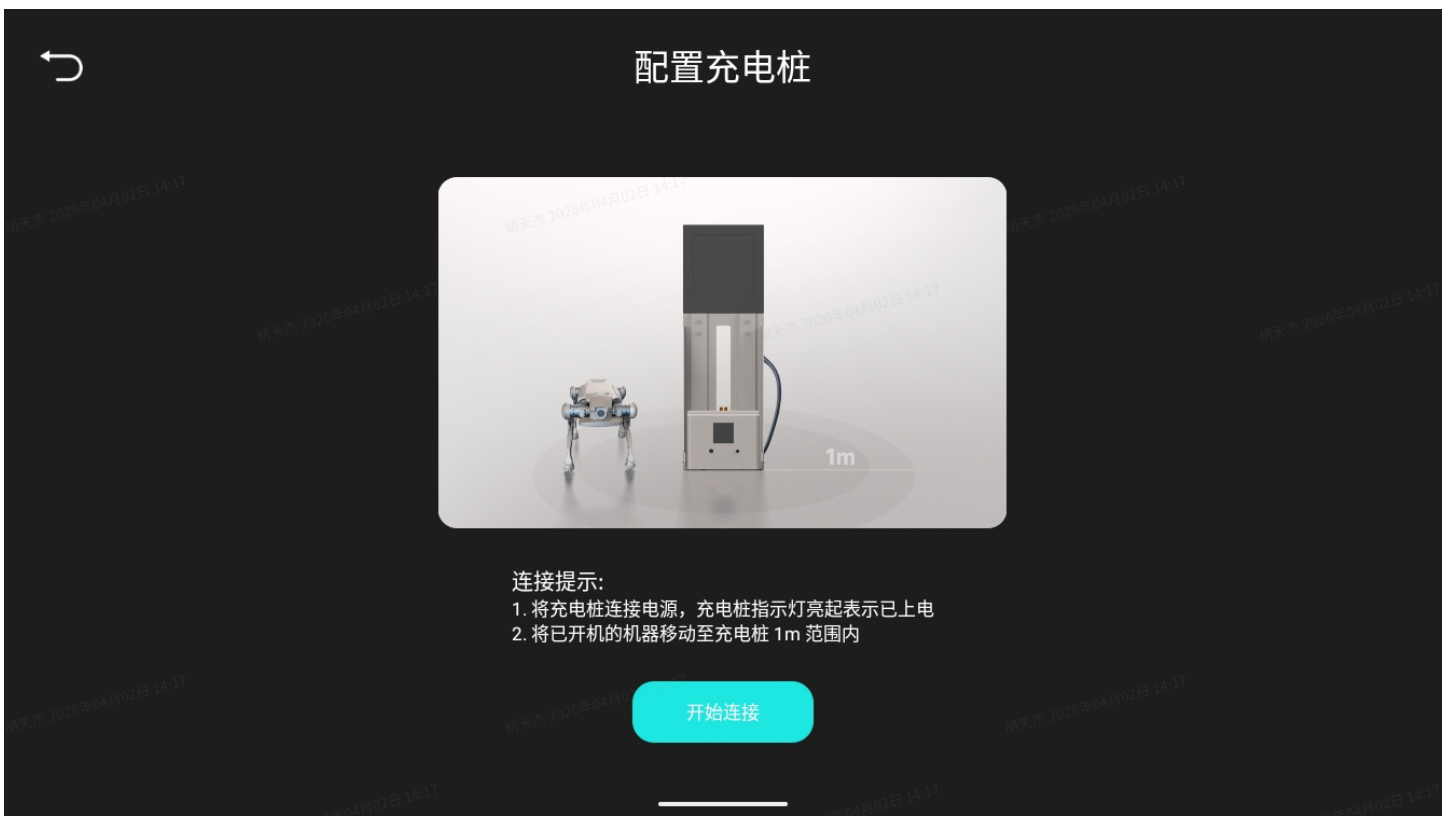
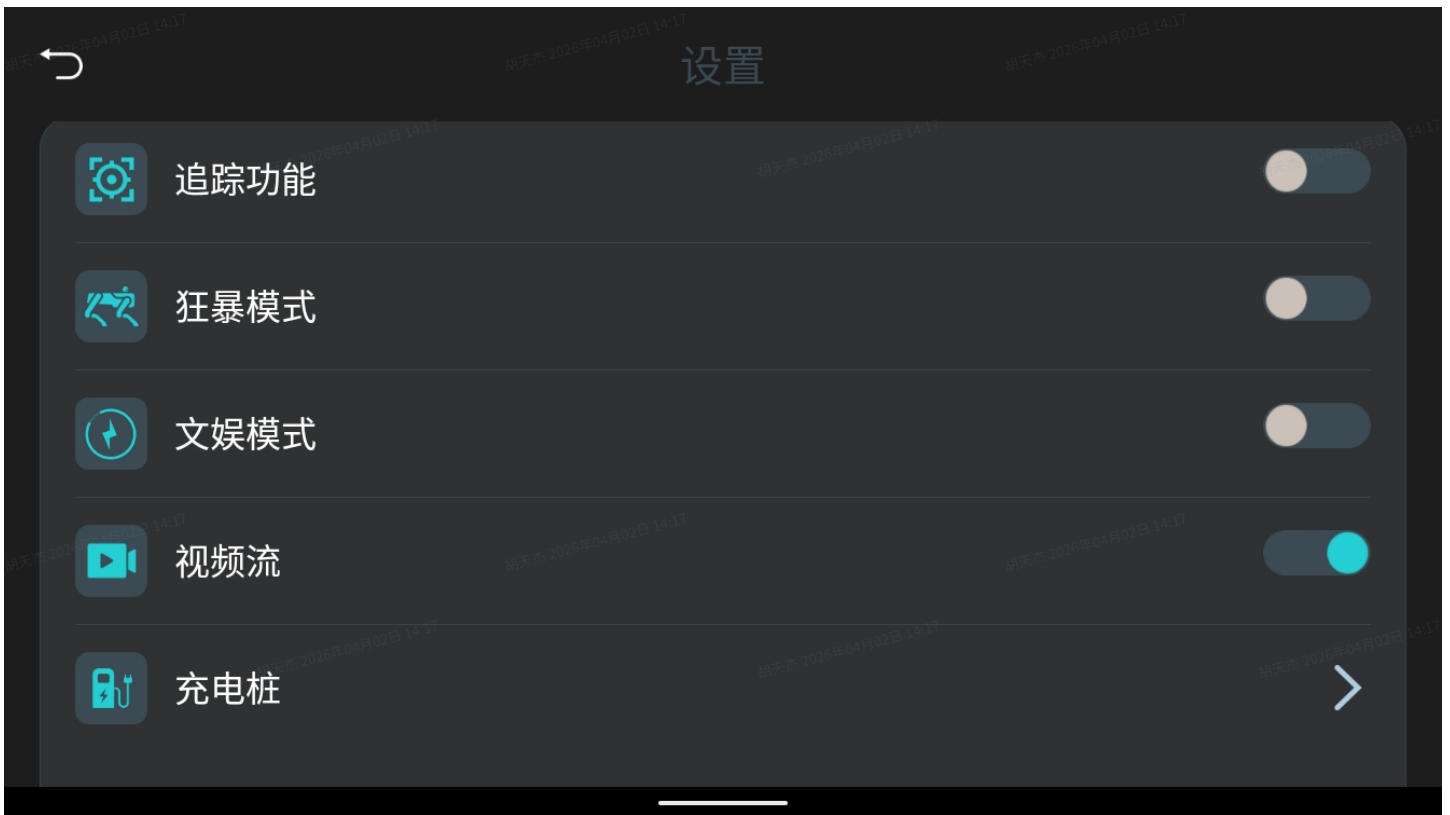
2、小狗遥控器给充电桩蓝牙配网

用小狗出厂自带的遥控器通过蓝牙给充电桩配网，要求小狗APP，版本 $\geq 2.4.7$ 。

1) 打开app，遥控器连上小狗蓝牙，点击左上角设置按钮；



2) 进入充电桩选项



3) 选择充电桩并进行配置，遥控器会将狗的蓝牙MAC给充电桩

配置充电桩



M1_dock_00000...

配置充电桩



正在搜索机器狗: XG01-589345...

4) 充电桩蓝牙连接上狗后，充电桩绿灯由快闪(250ms亮/250ms灭)变成了常亮。

3、小狗充电桩功能测试

3.1 将资料包发送到小狗RK3588S并解压

解压: `unzip charge_pile_xg_lib.zip`

```

root@robot:/home/robot# ls
charge_pile_xg_lib.zip  Documents  install  Pictures  Public  Templates  udp_client.sh  Videos
Desktop                Downloads  Music    pt        robot_launch_log  test_udp_receiver.py  udp_gui_client.py
root@robot:/home/robot# unzip charge_pile_xg_lib.zip ← 解压
Archive: charge_pile_xg_lib.zip
  creating: charge_pile_xg_lib/
  inflating: charge_pile_xg_lib/xg_charge_pile_sdk.zip ← 充电桩SDK和mavlink协议
  inflating: charge_pile_xg_lib/mavlink.zip
  inflating: charge_pile_xg_lib/charge_pile_board_main_86dad97_v0.0.3.bin ← 充电桩MCU固件
  creating: charge_pile_xg_lib/dog_send_three_states/
  inflating: charge_pile_xg_lib/dog_send_three_states/dog_status_unknown ← 狗下发的三种状态
  inflating: charge_pile_xg_lib/dog_send_three_states/dog_returning
  inflating: charge_pile_xg_lib/dog_send_three_states/dog_lying_down
  creating: charge_pile_xg_lib/charge_pile_ota/
  inflating: charge_pile_xg_lib/charge_pile_ota/mcu_upgrade ← 充电桩OTA脚本
root@robot:/home/robot# ls
charge_pile_xg_lib  Desktop  Downloads  Music  pt  robot_launch_log  test_udp_receiver.py  udp_gui_client.py
charge_pile_xg_lib.zip  Documents  install  Pictures  Public  Templates  udp_client.sh  Videos
root@robot:/home/robot# cd charge_pile_xg_lib/
root@robot:/home/robot/charge_pile_xg_lib# ls ← 解压后文件夹内部的文件
charge_pile_board_main_86dad97_v0.0.3.bin  charge_pile_ota  dog_send_three_states  mavlink.zip  xg_charge_pile_sdk.zip

```

3.2 测试充电桩给电池充电功能

```
sudo chmod +x dog_lying_down
```

```
sudo ./dog_lying_down
```

```

root@robot:/home/robot/charge_pile_xg_lib# ls
charge_pile_board_main_86dad97_v0.0.3.bin  charge_pile_ota  dog_send_three_states  mavlink.zip  xg_charge_pile_sdk.zip
root@robot:/home/robot/charge_pile_xg_lib# cd dog_send_three_states/
root@robot:/home/robot/charge_pile_xg_lib/dog_send_three_states# ls
dog_lying_down  dog_returning  dog_status_unknown
root@robot:/home/robot/charge_pile_xg_lib/dog_send_three_states# sudo chmod +x dog_lying_down ← 给予可执行权限
root@robot:/home/robot/charge_pile_xg_lib/dog_send_three_states# ls
dog_lying_down  dog_returning  dog_status_unknown
root@robot:/home/robot/charge_pile_xg_lib/dog_send_three_states# sudo ./dog_lying_down ← 运行可执行脚本

```

结果是充电桩绿灯会慢闪亮(500ms亮/500ms灭)，小狗上电池一般会闪烁，同时运行的dog_lying_down脚本会打印充电桩实时状态。

3.3 测试充电桩环境光传感器和补光灯功能

1) 将充电桩的环境光传感器挡住；

2) rk3588发送下面两条命令

```
sudo chmod +x dog_returning
```

```
sudo ./dog_returning
```

```

root@robot:/home/robot/charge_pile_xg_lib/dog_send_three_states# sudo chmod +x dog_returning
root@robot:/home/robot/charge_pile_xg_lib/dog_send_three_states# ls
dog_lying_down  dog_returning  dog_status_unknown
root@robot:/home/robot/charge_pile_xg_lib/dog_send_three_states# sudo ./dog_returning

```

结果是大小二维码补光灯会亮，同时运行的dog_returning脚本会打印充电桩实时状态。

3.4 测试充电桩OTA功能

1) 进入RK3588S，将使用ttyUSB0串口进程停止

`sudo fuser /dev/ttyUSB0` # 使用 fuser 确认，没有输出,表示没使用
若有输出，将此进程stop掉。

2) 查看当前充电桩固件运行版本

`sudo chmod +x mcu_upgrade`

```
sudo ./mcu_upgrade -d /dev/ttyUSB0 -s -c
```

```
root@robot:/home/robot/charge_pile_xg_lib/charge_pile_ota# ls
mcu_upgrade
root@robot:/home/robot/charge_pile_xg_lib/charge_pile_ota# sudo chmod +x mcu_upgrade
root@robot:/home/robot/charge_pile_xg_lib/charge_pile_ota# ls
mcu_upgrade
root@robot:/home/robot/charge_pile_xg_lib/charge_pile_ota# sudo ./mcu_upgrade -d /dev/ttyUSB0 -s -c
```

3) 给充电桩OTA

`sudo ./mcu_upgrade -d /dev/ttyUSB0 -f ../charge_pile_board_main_86dad97_v0.0.3.bin -c`

```
=====
get app b info
crc: 69d19c54
len: 16236
version: 0.0.1
date: 2025-12-01 20:57:34
commit_id: 8c317c6
ab: 801463d
=====

=====
get boot info
crc: b7437ffa
len: 10920
version: 0.0.1
date: 2025-12-01 20:57:34
commit_id: 8c317c6
ab: 0
=====
RUNNING APP B, UPGRADE APP A
erase app success
[#####] 100.00%
upgrade app success
firmware end
upgrade app end success
erase boot
erase boot success
boot write success
upgrade boot from end success
[all] success (success)
root@firefly:/home/firefly# ./mcu_upgrade -d /dev/ttyUSB0 -s -c
```

当前显示RUNNING APP B, UPGRADE APP A，升级后会显示100%精度条和下面success字样。

4) 充电桩断电，确认充电桩固件是否正常跳转到刚升级的固件，发送查看充电桩固件版本命令即可

`sudo ./mcu_upgrade -d /dev/ttyUSB0 -s -c`

5) 复测3.2充电桩充电和3.3环境光传感器和补光灯功能。

4、修改记录和资料包

参见小狗充电桩4.修改记录和资料包。

